Classification Report

YOLOv8n vehicle detection | Held-out test split | Prepared 03 November 2026

67.0%	52.2%	57.3%	35.6%
PRECISION	RECALL	mAP @ IoU 0.50	mAP @ IoU 0.50:0.95

Executive summary

The selected model is an Ultralytics YOLOv8n detector fine-tuned to locate and classify cars, motorcycles, buses, and trucks. On the held-out BDD100K-derived test split, it achieved 57.3% mAP@50 and 35.6% mAP@50:95. Overall recall was 52.2%, indicating that many labeled objects were not detected at the reported operating point.

This is an object-detection report. The class-wise precision, recall, and F1 values describe box detections matched to labels by class and location. Ordinary single-label accuracy is not an appropriate summary for this task.

Evaluation set and checkpoint

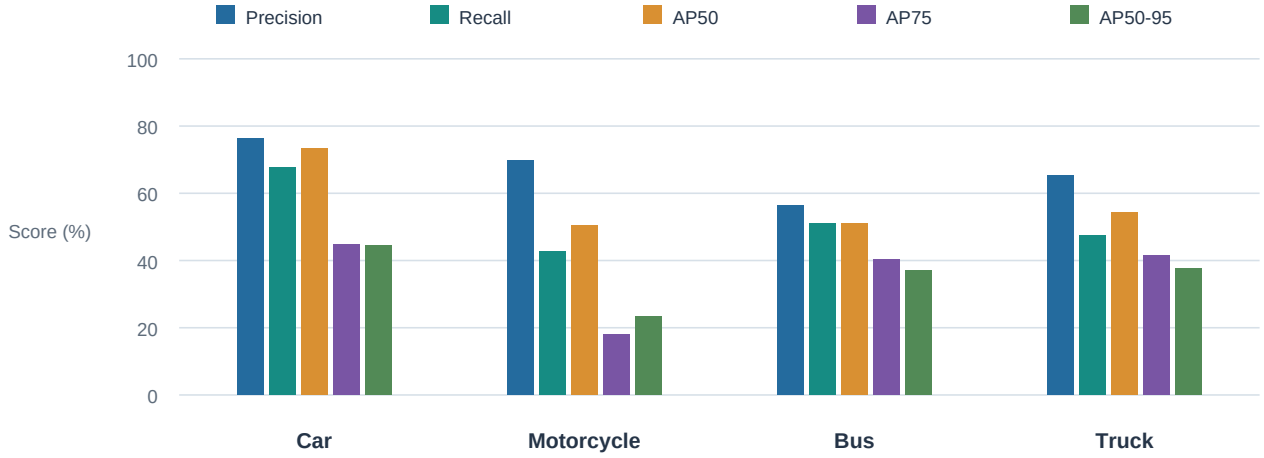
Item	Recorded value
Checkpoint	training/runs/traffic_finetune_from43_to60_20261101/weights/best.pt
Task / classes	Object detection: car, motorcycle, bus, truck
Held-out test split	2,610 images; 29,970 labeled object instances
Evaluation runtime	151.8 seconds on CUDA; this is total evaluation duration, not per-image latency

Dataset split sizes: 18,862 training images, 2,037 validation images, and 2,610 test images (23,509 total). The test split was used for final reporting and should not be used for further model or threshold tuning.

Per-class detection report

Support is the number of labeled object instances. 'Images' counts test images containing at least one instance of that class; a single image may contribute to multiple rows.

Class	Images	Objects	Precision	Recall	F1	AP@50	AP@75	AP@50:95
Car	2,588	26,903	76.2%	67.7%	71.7%	73.5%	44.7%	44.6%
Motorcycle	607	803	69.9%	42.7%	53.0%	50.3%	18.0%	23.2%
Bus	574	764	56.4%	51.0%	53.6%	51.0%	40.3%	36.9%
Truck	974	1,500	65.3%	47.5%	55.0%	54.3%	41.5%	37.5%
Overall	2,610	29,970	67.0%	52.2%	58.7%	57.3%	-	35.6%



Reading the results

Cars have the strongest AP@50 (73.5%) and recall (67.7%). Motorcycles have the lowest recall (42.7%) and mAP@50:95 (23.2%), consistent with much lower training and test support. Bus precision is the lowest (56.4%). These patterns suggest class-specific errors are hidden by the aggregate score.

F1 values are the harmonic mean of precision and recall at the evaluator's operating point. Overall F1 is calculated from the aggregate precision and recall and can differ slightly from an average of class F1 values.

Model and training configuration

The current checkpoint was produced by transfer learning: a previous VegMitra YOLOv8n checkpoint, itself initialized from COCO-pretrained weights, was continued from cumulative epoch 43 through epoch 60. The final continuation resumed from its own saved run state. `best.pt` was selected using validation fitness.

Recorded run parameters

Parameter	Recorded setting	Meaning / note
Architecture / task	YOLOv8n; detect	One-stage detector, four target classes
Checkpoint provenance	Previous VegMitra best -> continuation from epoch 43	COCO-pretrained ancestry; not trained from random initialization
Continuation duration	17 additional epochs; cumulative epoch 60	Recorded in run args/results
Input image size	640 x 640	Training and validation image size
Batch size	4	Images per training batch
Optimizer	AdamW	Adaptive gradient optimizer with decoupled weight decay
Learning rate	lr0 0.0002; lrf 0.1	Scheduled toward 10% of initial LR
Momentum / decay	0.937 / 0.0005	Recorded momentum and weight decay
Warmup	3 epochs; momentum 0.8; bias LR 0.1	Warmup settings recorded in args.yaml
Seed / determinism	42 / enabled	Reproducibility settings
Hardware / workers	CUDA device 0; 8 workers	Evaluation JSON records CUDA; run log identifies laptop GPU
AMP / validation	AMP enabled; validation enabled	Mixed precision and validation during training
Patience	18	Configured patience; continuation completed at target epoch 60
Effective batch / multi-scale	nbs 64; multi-scale 0	Nominal batch size for loss scaling; multi-scale disabled
Schedule / determinism	cosine LR off; deterministic on	Recorded run flags

Augmentation and loss parameters

Parameter	Value	Parameter	Value
HSV hue / saturation / value	0.015 / 0.7 / 0.4	Horizontal flip	0.5
Translation / scale	0.1 / 0.5	Rotation / vertical flip	0 / 0
Mosaic / close mosaic	1.0 / 2 epochs	MixUp / CutMix	0 / 0
Box / class / DFL loss gains	7.5 / 0.5 / 1.5	Weight decay	0.0005
IoU / max detections	0.7 / 300	Single-class mode	off
Shear / perspective	0 / 0	HSV BGR / copy-paste	0 / 0
Auto-augment / erasing	RandAugment / 0.4	Warmup bias LR	0.1

Recorded software: Python 3.13.15, PyTorch 2.15.0+cu132, Ultralytics 8.4.195; CUDA device 0 (NVIDIA GeForce RTX 5050 Laptop GPU per project run notes). Training curves and confusion matrices are saved in the run folder. The config-driven `training/train.py` script starts a new run with different settings; it is not a command to reproduce this exact continuation.

Dataset composition

The detection dataset is derived from BDD100K annotations converted to normalized YOLO boxes. Only car, motorcycle (BDD category `motor`), bus, and truck are included. Cars dominate the labels; motorcycles are the minority class.

Dataset YAML: `training/dataset.yaml`; configured data root: local expanded BDD100K-derived dataset
Dataset files are not included in the public source repository.

Split	Images	Car boxes	Motorcycle boxes	Bus boxes	Truck boxes	Total boxes
Train	18,862	193,754	3,002	7,642	10,002	214,400
Validation	2,037	21,059	358	532	1,004	22,953
Test	2,610	26,903	803	764	1,500	29,970
Total	23,509	241,716	4,163	8,938	12,506	267,323

Evaluation procedure

The project evaluator loaded the selected `best.pt` checkpoint and invoked Ultralytics validation on `training/dataset.yaml` with `split=test`. It did not pass explicit `imgsz`, `conf`, or `iou` overrides; therefore those values follow the installed Ultralytics version's validation defaults/model settings. The saved evaluation JSON records checkpoint, split, metrics, class support, duration, and CUDA device, but does not record every resolved evaluator default. The run's training arguments record 640-pixel images and IoU 0.7 for training-time validation.

Reproduction command: `python training/evaluate.py --model training/runs/traffic_finetune_from43_to60_20261101/weights/best.pt --split test`.
The evaluator writes the JSON and per-class CSV used for this report.

Metric guide

Metric	Definition and use
Precision	Correct matched detections divided by all predicted detections; lower precision means more false detections.
Recall	Matched labeled objects divided by all labeled objects; lower recall means more misses.
IoU	Intersection-over-union of a predicted box and its ground-truth box; measures localization overlap.
AP@50 / AP@75	Per-class area under precision-recall curve using IoU match thresholds of 0.50 / 0.75.
mAP@50:95	Mean class AP averaged over IoU thresholds 0.50, 0.55, ..., 0.95; more demanding localization summary.
F1	2 x precision x recall / (precision + recall), at one operating point; not accuracy.

Results are tied to this checkpoint, dataset split, and evaluator version. A controlled comparison against the original COCO model on the same BDD100K-derived test set has not been reported.

What the results mean

The test mAP@50 of 57.3% summarizes class-averaged precision-recall performance when a detection is considered matched at IoU 0.50. The stricter mAP@50:95 of 35.6% indicates that box placement is less consistent at higher overlap thresholds. Overall recall of 52.2% means roughly half of labeled instances were matched at the reported operating point. This is material for queue counting, where missed objects can reduce estimated demand.

The class support is imbalanced: 26,903 car boxes versus 803 motorcycle boxes in test. Motorcycle and bus metrics should therefore be examined separately instead of relying only on aggregate results.

System boundaries

YOLO detects vehicle boxes; BoT-SORT tracks detections; geometry estimates approach and movement; rolling averages and density formulas feed a programmed green-road selector. The detector does not learn traffic phases. The separate Decision Tree / Random Forest / AdaBoost / Gradient Boosting experiment uses 600 synthetic rows with density-derived labels and is not integrated into the controller.

The current detector has no trained ambulance, fire-truck, or police classes. Emergency scenarios in the MATLAB application are simulated. Detection results on BDD100K images do not prove accurate per-approach counts, real-time throughput, safe signal timings, or field readiness.

Recommended next evaluation

Use manually labeled local intersection videos to measure per-class detection error, per-approach count MAE/RMSE, track duplication and fragmentation, decision stability, latency/FPS, and performance under occlusion and lighting changes. Tune confidence thresholds on validation footage; preserve the test split for final reporting.

Source artifacts

Primary metric source: `training/results/evaluation_metrics_test.json` and `training/results/classification_report_test.csv`. Checkpoint parameters: `training/runs/traffic_finetune_from43_to60_20261101/args.yaml`. Dataset specification: `training/dataset.yaml`. Evaluation implementation: `training/evaluate.py`. Project context and per-class validation results: `docs/PROJECT_CONTEXT.md` and `docs/MODEL_EVALUATION.md`.

Report prepared from the saved epoch-60 checkpoint evaluation. No model training or reevaluation was performed to generate this document.